



How big data will change our future

Data defines our lives in increasingly breathtaking ways. In its raw unorganized form, data is found in the alphabet, numbers and symbols we use daily. Collectively, our data trails reveal our behaviors, purchasing habits and social interactions. Companies wish to harness the expanding clouds of data, attempting to pinpoint us with better products, services and marketing offers. Medical organizations have massive data sets that track health records, population changes and more. Scientists and government officials face this data challenge in climate change, biological research, financial information, national security tracking and other areas.

These data sets have become so unwieldy and awkward that they've earned the term of 'big data'. Big data is an ever-present problem that pushes the limits of current data management tools.

Big data also suggests big impact. In short, companies are now collecting so much information about our activities that companies who can analyze and capitalize on the vast data patterns gain a tremendous market advantage. Wealth management firm Merrill Lynch recently reported that big data would lead the coming decade of revolutionary businesses, services and Wall Street deals.

Frankly, big data is not a new revolution. It has been occurring since the invention of the printing press, which

started the first indexing problem. Suddenly, society had to address how a reader could find a specific topic or word within a treasured book. The solution was to assign the task to the most educated humans, who would spend almost their entire lives generating detailed numerical or alphabetical listings.

In more modern times, Yahoo! addressed the early web problem in exactly the same way – it assigned people to generate indices. Then, along came Google with its computer algorithm that could automatically sift through the big data set of web pages. Through multiple iterations, Google discovered how to efficiently map and then monetize the network graph that underlay the world wide web of pages.

We're now witnessing an explosion of social information. The current leader is Facebook, a social networking service that allows over 500 million users to exchange personal information continuously. The unit of big data has shifted from pages to people. People linking to other people, building a new network layer to be analysed and monetised.

A third class of big data is emerging that builds on these networks of pages and people. The new unit is, coarsely, the product.

The different and complex associations between products raise new questions and expectations for big data management. If a person reads a book

review online, shouldn't that review automatically find and ideally present the reader with an opportunity to purchase an e-book copy, along with a connection to recent reviews, the author's upcoming book tour and a discussion of the book by a renowned writer? As a business manager plans a trip to London, what if the travel article she's reading also suggests similar articles, automatically extracts the restaurants, hotels and events across the articles (all products), and then helps her assemble those articles into an itinerary? Or, when a couple is enjoying good wine at a restaurant, the bottle can point them to a list of foods typically paired with it, so that they may buy a second bottle for an event next weekend.

These cases foretell a world of big data, in which you can weave your own story, tailor your own collections and become a producer of your life with nothing more than your own ability to curate what you like. As you build your own interconnected world, you'll point to other web pages, people and even products – and they all may link back to you.

This future of big data has already begun, and realistically, the creation of small parts of this data set is being done first by hand. Slowly. And then, more parts will be automated until it seems almost magical, and like other big ideas, you'll wonder how you ever lived without it. 



William Cockayne and Tamara Carleton

William Cockayne, PhD, is a seasoned entrepreneur in Silicon Valley and a Consulting Associate Professor at Stanford University.

Tamara Carleton, PhD, is president and founder of Innovation Leadership Board and also a Fellow at the Foundation for Enterprise Development.